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Devices, systems, and methods of manufacturing fluid-cooled ultrasonic surgical instruments

Abstract

An ultrasonic surgical system includes a waveguide defining first and second longitudinal lumens extending through at least a portion of a length of the waveguide. A distal transverse lumen is defined within the waveguide transversely through at least a portion of the waveguide to intersect and interconnect the first and second longitudinal lumens. First and second proximal transverse lumens extend from first and second proximal transverse apertures within the waveguide transversely through portions of the waveguide to intersect the first and second longitudinal lumens, respectively. Inflow and outflow conduits are fluidly coupled with the first and second proximal transverse apertures, respectively, to enable the inflow of fluid into the first longitudinal lumen and the outflow of fluid from the second longitudinal lumen, respectively.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of, and priority to, U.S. Provisional Patent Application No. 63/134,268 filed on Jan. 6, 2021, the entire contents of which are hereby incorporated herein by reference.

FIELD
(1) The present disclosure relates to ultrasonic surgical instruments and, more particularly, to devices, systems, and methods of manufacturing fluid-cooled ultrasonic surgical instruments.
BACKGROUND
(2) Ultrasonic surgical instruments and systems utilize ultrasonic energy, i.e., mechanical vibration energy transmitted at ultrasonic frequencies, to treat, e.g., seal and/or transect, tissue. Ultrasonic surgical instruments typically include a waveguide having a transducer coupled to a proximal end portion of the waveguide and an end effector disposed at a distal end portion of the waveguide. The waveguide transmits the ultrasonic energy produced by the transducer to the end effector for treating tissue at the end effector. The end effector may include a blade, hook, ball, etc. and/or other features such as a clamping mechanism for clamping tissue against the end effector and/or to facilitate manipulating tissue. During use, the waveguide and/or end effector of an ultrasonic surgical instrument can reach temperatures greater than 200° C. or even 300° C.
SUMMARY
(3) As used herein, the term “distal” refers to the portion that is described which is further from an operator (whether a human surgeon or a surgical robot), while the term “proximal” refers to the portion that is being described which is closer to the operator. Terms including “generally,” “about,” “substantially,” and the like, as utilized herein, are meant to encompass variations, e.g., manufacturing tolerances, material tolerances, use and environmental tolerances, measurement variations, and/or other variations, up to and including plus or minus 10 percent. Further, any or all of the aspects described herein, to the extent consistent, may be used in conjunction with any or all of the other aspects described herein.
(4) Provided in accordance with aspects of the present disclosure is an ultrasonic surgical system including an ultrasonic waveguide body defining first and second longitudinal lumens extending through at least a portion of a length of the ultrasonic waveguide body. A distal transverse lumen is defined within the ultrasonic waveguide body transversely through at least a portion of the ultrasonic waveguide body to intersect and interconnect the first and second longitudinal lumens. A first proximal transverse lumen extends from a first proximal transverse aperture within the ultrasonic waveguide body transversely through a portion of the ultrasonic waveguide body to intersect the first longitudinal lumen. A second proximal transverse lumen extends from a second proximal transverse aperture within the ultrasonic waveguide body transversely through a portion of the ultrasonic waveguide body to intersect the second longitudinal lumen. Inflow and outflow conduits are fluidly coupled with the first and second proximal transverse apertures, respectively, to enable the inflow of fluid into the first longitudinal lumen and the outflow of fluid from the second longitudinal lumen, respectively.
(5) In an aspect of the present disclosure, the distal transverse lumen extends from a distal transverse aperture and a distal transverse aperture plug closes the distal transverse aperture to

inhibit escape of fluid therethrough.
(6) In an aspect of the present disclosure, the first and second longitudinal lumens extend proximally from first and second distal face apertures defined within a distal face of the ultrasonic waveguide body. In such aspects, at least one distal face aperture plug closes the first and second distal face apertures to inhibit escape of fluid therethrough.
(7) In another aspect of the present disclosure, the first and second proximal transverse apertures are defined on opposed sides of the ultrasonic waveguide body. Alternatively, the first and second proximal transverse apertures may be defined on the same side of the ultrasonic waveguide body.
(8) In still another aspect of the present disclosure, first and second proximal transverse aperture plugs form a seal between the inflow and outflow conduits and the first and second proximal transverse apertures, respectively.
(9) In yet another aspect of the present disclosure, the ultrasonic surgical system further includes a proximal waveguide body adapted to connect to an ultrasonic transducer. In such aspects, the ultrasonic waveguide body is a distal waveguide body including a blade and extends distally from the proximal waveguide body. The distal waveguide body may be releasably engagable with the proximal waveguide body or permanently affixed (or formed with) the proximal waveguide body. Engagement of the proximal and distal waveguide bodies, in aspects where such engagement is provided, may close proximal ends of the first and second longitudinal lumens.
(10) In still yet another aspect of the present disclosure, the waveguide body includes a base and a blade extending distally from the base. The blade may define opposed narrow surfaces and opposed broad surfaces. At least a portion of the distal transverse lumen may be disposed within 10% of a length of the blade from a distal end of the blade.
(11) In another aspect of the present disclosure, the ultrasonic surgical system further includes a cooling system configured to pump cooling fluid through the inflow conduit into the first longitudinal lumen and/or pump cooling fluid through the second longitudinal lumen into the outflow conduit.
(12) In another aspect of the present disclosure, the ultrasonic surgical system further includes a housing supporting the cooling system and an elongated assembly extending distally from the housing. The elongated assembly includes the ultrasonic waveguide body.
(13) In yet another aspect of the present disclosure, the housing further supports an ultrasonic transducer configured to produce ultrasonic energy for transmission along the ultrasonic waveguide body. The housing may further still support an ultrasonic generator configured to produce an ultrasonic drive signal for driving the ultrasonic transducer, and/or a battery configured to power the ultrasonic generator.
(14) In another aspect of the present disclosure, the distal transverse lumen is at least partially formed via a distal cap defining a distal tip of the distal waveguide body.
(15) A method of manufacturing an ultrasonic surgical system provided in accordance with the present disclosure includes forming first and second longitudinal lumens through at least a portion of a length of an ultrasonic waveguide body, forming a distal transverse lumen transversely through at least a portion of the ultrasonic waveguide body to intersect and interconnect the first and second longitudinal lumens, forming first and second proximal transverse lumens transversely through a portion of the ultrasonic waveguide body to intersect the first and second longitudinal lumens, respectively, plugging apertures formed from the forming of the first and second longitudinal lumens, and plugging an aperture formed from the forming of the distal transverse lumen.
(16) In an aspect of the present disclosure, the method further includes fluidly connecting an inflow conduit with the first proximal transverse aperture and/or fluidly connecting an outflow conduit with the second proximal transverse aperture.
(17) In another aspect of the present disclosure, the method further includes attaching the ultrasonic waveguide body, as a distal waveguide body, to a proximal waveguide body. Such attaching may close proximal ends of the first and second longitudinal lumens. The distal waveguide body may be

attached to the proximal waveguide body via inertial friction welding or in any other suitable manner.
(18) In still another aspect of the present disclosure, the method further includes coupling the first and second proximal transverse lumens with inflow and outflow conduits, respectively, associated with a cooling system.
(19) In another aspect of the present disclosure, forming the distal transverse lumen includes securing a cap to a distal end of the ultrasonic waveguide body to define a distal tip of the ultrasonic waveguide body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above and other aspects and features of the present disclosure will become more apparent in light of the following detailed description when taken in conjunction with the accompanying drawings wherein like reference numerals identify similar or identical elements.
(2) FIG. **1** is a perspective view of an ultrasonic surgical instrument provided in accordance with the present disclosure;
(3) FIG. **2** is a side, cut-away view of a proximal portion of the ultrasonic surgical instrument of FIG. **1**;
(4) FIG. **3** is a schematic illustration of a robotic surgical system provided in accordance with the present disclosure;
(5) FIG. **4**A is an enlarged, perspective view of a distal portion of a waveguide of the ultrasonic surgical instrument of FIG. **1**;
(6) FIG. **4**B is a longitudinal, cross-sectional view of a distal body of the waveguide of FIG. **4**A illustrating an internal flow path formed therein;
(7) FIG. **5** is a perspective view of the distal portion of the waveguide of FIG. **4**A assembled with inflow and return conduits connected thereto and the fluid flow path sealed closed;
(8) FIGS. **6**A and **6**B are perspective and longitudinal, cross-sectional views, respectively, of a distal portion of another waveguide provided in accordance with the present disclosure;
(9) FIG. **7** is a longitudinal, cross-sectional view of a distal portion of still another waveguide provided in accordance with the present disclosure;
(10) FIGS. **8**A and **8**B are perspective and longitudinal, cross-sectional views, respectively, of a distal portion of yet another waveguide provided in accordance with the present disclosure; and
(11) FIGS. **9** and **10** are longitudinal, cross-sectional views of proximal portions of still yet other waveguides provided in accordance with the present disclosure.
DETAILED DESCRIPTION
(12) Turning to FIGS. **1** and **2**, an ultrasonic surgical instrument provided in accordance with aspects of the present disclosure is generally identified by reference numeral **10**. Instrument **10** is a fully cordless instrument that incorporates an on-board cooling system in addition to an on-board power source, e.g., battery, and an ultrasonic generator and transducer. However, it is also contemplated that instrument **10** be configured as a corded instrument, e.g., wherein instrument **10** is configured to connect to a remote cooling system by way of one or more fluid lines and to a remote ultrasonic generator (separate from or integrated with the remote cooling system) by way of a cable; or as a partially-corded instrument, e.g., wherein instrument **10** includes an on-board power source and ultrasonic generator and is configured to connect to a remote cooling system by way of one or more fluid lines, or wherein instrument **10** incorporates an on-board cooling system and is configured to connect to a remote ultrasonic generator. Likewise, other style ultrasonic instruments are also contemplated such as, for example, pencil-style instruments, hemostat-style instruments, etc. Thus, although particular aspects and features of instrument **10** are detailed below, it is

understood that the aspects and features of the present disclosure are equally applicable for use with any other suitable ultrasonic surgical instrument or system, e.g., robotic surgical system **1000** (FIG. **3**). Other suitable instruments for use in accordance with the present disclosure including remote or on-board cooling systems are described, for example, in Patent Application Pub. No. US 2019/0247073, titled “REMOVABLE FLUID RESERVOIR AND ULTRASONIC SURGICAL INSTRUMENT INCLUDING THE SAME” and filed on Feb. 13, 2018, and Patent Application Pub. No. US 2017/0281215, titled “DEVICES, SYSTEMS, AND METHODS FOR COOLING A SURGICAL INSTRUMENT” and filed on Mar. 18, 2017, the entire contents of each of which are hereby incorporated herein by reference.

(13) Instrument **10** generally includes a handle assembly **100**, an elongated assembly **200** that extends distally from handle assembly **100**, a transducer and generator assembly (“TAG”) **300** configured for releasable engagement with handle assembly **100**, and a battery **400** configured for removable receipt within handle assembly **100**. Elongated assembly **200** may be integral with handle assembly **100** or may be releasably engagable with handle assembly **100**.

(14) Handle assembly **100** includes a housing **110**, a cooling system **120**, a switch assembly **140**, a generator dock **150**, a battery dock **160**, a flex circuit assembly **170** (including flex circuit portions **182**, **184**), and a clamp trigger **190**.

(15) Housing **110** of handle assembly **100** defines a body portion **112** defining a longitudinal axis and a fixed handle portion **114** extending from body portion **112** at an oblique angle relative to the longitudinal axis of body portion **112** (although fixed handle portion **114** may alternatively extend perpendicularly relative to the longitudinal axis of body portion **112**). Body portion **112** of housing **110** is configured to receive a proximal portion of elongated assembly **200** in operable engagement with clamp trigger **190** such that actuation of clamp trigger **190** manipulates end effector assembly **280** of elongated assembly **200**. When engaged with body portion **112** of housing **110**, elongated assembly **200** is aligned on the longitudinal axis of body portion **112**. Body portion **112** of housing **110** is also configured to support TAG **300** thereon with transducer **320** of TAG **300** mechanically coupled with waveguide **230** of elongated assembly **200**, e.g., via a threaded connection, latching, or in any other suitable manner, and both aligned on the longitudinal axis of body portion **112** of housing **110**. Generator **340** of TAG **300** is electrically coupled with generator dock **150** of housing **110** when TAG **300** is engaged with body portion **112** of housing **110**. Fixed handle portion **114** of housing **110** defines an internal compartment **116** configured to removably receive battery **400** therein and a hinged door **118** configured to enclose battery **400** within internal component **116**.

(16) Cooling system **120** includes one or more fluid pumps **122** and, in some aspects, a fluid reservoir **124**, although fluid reservoir **124** may be omitted in other configurations. Cooling system **120** further includes associated tubing **126** operably interconnecting fluid pump **122**, fluid reservoir **124**, and inflow and return conduits **128***a*, **128***b* (or interconnecting fluid pump **122** and inflow and return conduits **128***a*, **128***b*, in aspects where fluid reservoir **124** is omitted) for pumping cooling fluid to and returning cooling fluid from elongated assembly **200**. Inflow and return conduits **128***a*, **128***b* may extend along and/or through elongated assembly **200**, ultimately entering waveguide **230** thereof to permit circulation of cooling fluid through blade **282** of end effector assembly **280** of elongated assembly **200**, as described in greater detail below.

(17) The one or more fluid pumps **122** of cooling system **120** are supported within body portion **112** of housing **110**. For example, a fluid pump **122** may be supported on either or both sides of body portion **112** of housing **110** at radially-spaced positions relative to the longitudinal axis of body portion **112** of housing **110**. In such a configuration, fluid pumps **122** may define relatively thin, elongate configurations such that sufficient space is defined between the pumps **122** and/or the pump **122** and housing **110** to permit passage of transducer **320** of TAG **300** therebetween (in alignment on the longitudinal axis) while requiring minimal, if any, increase in the overall with dimension of body portion **112** of housing **110** to accommodate pump(s) **122**.

(18) A connection interface **130** of cooling system **120** for enabling power and/or control signals to

be transmitted to pump(s) **122** is positioned so as not to interfere with TAG **300**. Further, a connector **132**, e.g., lead wire, cable, flex circuit, or other suitable connector, extends through body portion **112** of housing **110** to couple the connection interface **130** with flex circuit assembly **170** of handle assembly **100** to permit communication of power and control signals between pump(s) **122**, generator **340**, switch assembly **140**, and/or battery **400**. More specifically, pump(s) of cooling system **120** may be controlled via a controller of generator **340**, battery **400**, or a separate controller of cooling system **120**, e.g., within control box **130**. Regardless of the location and/or configuration, the controller is configured to control pump(s) **122** so as to maintain a flow of cooling fluid sufficient to cool blade **282** of end effector **280**, to activate and deactivate cooling in response to manual inputs, and/or to implement automatic cooling (for example, upon deactivation of the supply of energy). The one or more fluid pumps **122** may be piezoelectric microfluidic pumps or other microfluidic pumps such as micro-peristaltic pumps, syringe pumps, etc. Regardless of the particular pump configuration utilized, the one or more fluid pumps **122** may, in aspects, be configured to generate sufficient flow rate of cooling fluid so as to cool waveguide **230** of elongated assembly **200** from an initial temperature of about 300° C. to about 100° C. (or about 120° C.) to a cooled temperature of about 70° C. to about 0° C. (or less than about 60° C.) in from about 0.5 seconds to about 2.5 seconds (or in less than about 2 seconds). However, other temperatures and/or cooling times are also contemplated.
(19) Continuing with reference to FIGS. **1** and **2**, fluid reservoir **124** is disposed within housing **110** of handle assembly **100**. More specifically, fluid reservoir **124** is disposed within fixed handle portion **114** of housing **110** and is positioned between internal compartment **116** of fixed handle portion **114** and body portion **112** of housing **110**. Fluid reservoir **124** may define a cut-out **134** within which at least a portion of clamp trigger **190** extends upon actuation thereof. As such, fluid reservoir **124** does not interfere with actuation of clamp trigger **190**. It is also contemplated that fluid reservoir **124** be positioned in other locations, e.g., at a free end of fixed handle portion **114** such that battery **400** is disposed between fluid reservoir **124** and body portion **112**.
(20) Fluid reservoir **124** further includes a port **136** having an input **137***a* and an output **137***b*. Tubing **126** of cooling system **120** is coupled to input **137***a* and output **137***b* of fluid reservoir **124** to couple fluid reservoir **124** with fluid pump(s) **122** and inflow and return conduits **128***a*, **128***b*. More specifically, either or both of the ends **138***a*, **138***b* of the tubes of tubing **126** may extend through input **137***a* and output **137***b* and into fluid reservoir **124** such that the ends **138***a*, **138***b* of the tubes of tubing **126** are disposed at opposite sides, ends, or portions of fluid reservoir **124**, thereby maximizing the spacing therebetween. This configuration inhibits the hotter, returned cooling fluid from being immediately pumped back out of fluid reservoir **124**. Further, the ends **138***a*, **138***b* of the tubes of tubing **126** are positioned within fluid reservoir **124** relative to one another such that, regardless of the orientation of handle assembly **100**, any air in fluid reservoir **124** is inhibited from entering inflow conduit **128***a*.
(21) Switch assembly **140** of handle assembly **100** includes an energy activation button **142** operably positioned to electrically couple to flex circuit assembly **170**. Flex circuit assembly **170** electrically couples switch assembly **140**, battery **400**, and TAG **300** with one another. Thus, when energy activation button **142** is activated in an appropriate manner, power is supplied from battery **400** to TAG **300**. Energy activation button **142** may be configured for dual-mode activation such that, a first activation of energy activation button **142** drives TAG **300** in a “LOW” power mode, while a second, different activation of energy activation button **142** drives TAG **300** in a “HIGH” power mode. Other suitable activation configurations are also contemplated.
(22) Switch assembly **140** of handle assembly **100** further includes, in aspects, a pair of cooling activation buttons **144** operably positioned on either side of housing **110**. Flex circuit assembly **170** electrically couples connection interface **130** of cooling system **120** with switch assembly **140**, battery **400**, and TAG **300**. Thus, activation of either or both of cooling activation buttons **144** initiates cooling. In aspects, multiple activations and/or particular activation patterns of cooling

activation button(s) **144** may subsequently terminate cooling, switch between different cooling modes or programs, etc.

(23) Generator dock **150** is disposed on body portion **112** of housing **110** and is positioned to electrically couple to generator **340** of TAG **300** upon engagement of TAG **300** with housing **110**. Battery dock **160** is disposed within internal compartment **116** of fixed handle portion **114** of housing **110** and is positioned to electrically couple to battery **400** upon receipt of battery **400** within internal compartment **116**. Docks **150**, **160**, flex circuit assembly **170** (including flex circuit portions **182**, **184** thereof), and connector **132** of cooling system **120** electrically couple TAG **300**, switch assembly **140**, control box **130** of cooling system **120**, and battery **400** with one another to enable communication of power and/or control signals therebetween. Flex circuit portion **182**, more specifically, interconnects battery dock **160** with flex circuit assembly **170**. The flexible configuration of flex circuit portion **182** enables routing of flex circuit **182** about fluid reservoir **124**, which is disposed between flex circuit assembly **170** and battery dock **160**. Flex circuit portion **184**, on the other hand, electrically couples flex circuit assembly **170** with generator dock **150**.

(24) Referring still to FIGS. **1** and **2**, clamp trigger **190** of handle assembly **100** of instrument **10** extends from body portion **112** of housing **110** in opposing relation relative to fixed handle portion **114** of housing **110**. Clamp trigger **190** is pivotably coupled to body portion **112** of housing **110** and operably associated with elongated assembly **200** such that pivoting of clamp trigger **190** towards fixed handle portion **114** of housing **110** pivots jaw **284** of end effector assembly **280** of elongated assembly **200** from an open position to a clamping position for clamping tissue between jaw **284** and blade **282**, which extends distally from waveguide **230** of elongated assembly **200**.

(25) Elongated assembly **200** generally includes a sleeve assembly having an outer sleeve **210** and an inner sleeve (not shown) disposed within outer sleeve **210**, waveguide **230** extending through the inner sleeve (not shown), a drive assembly **250**, a rotation assembly **270** operably disposed about outer sleeve **210**, and end effector assembly **280** disposed at the distal end of the sleeve assembly. End effector assembly **280** includes, as noted above, blade **282** and jaw **284**, which is operably coupled to outer sleeve **210** such that translation of outer sleeve **210** pivots jaw **284** relative to blade **282** between the open and clamping positions. Drive assembly **250** operably couples a proximal portion of outer sleeve **210** with clamp trigger **190** such that actuation of clamp trigger **190** pivots jaw **284** relative to blade **282** between the open and clamping positions. In the above-detailed configuration, the inner sleeve is a support sleeve and the outer sleeve is a drive sleeve, although the opposite configuration is also contemplated, as are other suitable drive mechanisms for pivoting jaw **284** relative to blade **282**.

(26) Jaw **284** of end effector assembly **280** includes a more-rigid structural body **285***a* and a more-compliant jaw liner **285***b*. Structural body **285***a* is pivotably coupled to the inner sleeve of elongated assembly **200** and operably coupled with outer sleeve **210** of elongated assembly **200** to enable the above-detailed translation of outer sleeve **210** to impart pivotal motion of jaw **284** relative to blade **282** to clamp tissue between jaw liner **285***b* of jaw **284** and blade **282**. Jaw liner **285***b* is positioned to oppose blade **282** in the clamping position of jaw **284**.

(27) Waveguide **230** extends through the inner sleeve (not shown), includes blade **282** extending from the distal end thereof and includes a proximal end portion that is configured to operably couple to transducer **320**, e.g., via a threaded connection, latching, or in any other suitable manner. Inflow and return conduits **128***a*, **128***b*, in aspects, may extend from housing **110** at least partially along and/or through elongated assembly **200** before fluidly coupling with an interior flow path defined within waveguide **230** or may fluidly couple with waveguide **230** within housing **110**. Waveguide **230**, the interior flow path defined within waveguide **230**, and the connection of inflow and return conduits **128***a*, **128***b* to waveguide **230** are described in greater detail below.

(28) TAG **300** and battery **400** are each removable from handle assembly **100** to facilitate disposal of handle assembly **100** or to enable sterilization of handle assembly **100**. TAG **300** may be configured to withstand sterilization such that TAG **300** may be sterilized for repeated use. Battery

400, on the other hand, is configured to be aseptically transferred and retained within compartment **116** of fixed handle portion **114** of housing **110** of handle assembly **100** such that battery **400** may be repeatedly used without requiring sterilization thereof. Alternatively or additionally, battery **400** may be sterilized. In some configurations, TAG **300** (or a portion thereof, e.g., generator **340** or transducer **320**) may be integral with housing **110**.

(29) TAG **300** includes ultrasonic transducer **320** and generator **340**. A set of connectors **362** and corresponding rotational contacts **364** associated with generator **340** and ultrasonic transducer **320**, respectively, enable data and drive signals to be communicated from generator **340** to transducer **320**, e.g., the piezoelectric stack of transducer **320**, to drive transducer **320**. Battery **400** powers generator **340** to produce a drive signal, e.g., a high voltage AC signal, that is communicated to transducer **320**. Transducer **320** converts the signal into mechanical motion that is output along waveguide **230** to blade **282** of end effector assembly **280**. Transducer **320** further includes a rotation knob **380** disposed at a proximal end thereof to enable rotation of transducer **320** relative to generator **340** and handle assembly **100**. Rotation knob **380** may also facilitates operable coupling of transducer **320** with elongated assembly **200**.

(30) With reference to FIG. **3**, a robotic surgical system in accordance with the aspects and features of the present disclosure is shown generally identified by reference numeral **1000**. For the purposes herein, robotic surgical system **1000** is generally described. Aspects and features of robotic surgical system **1000** not germane to the understanding of the present disclosure are omitted to avoid obscuring the aspects and features of the present disclosure in unnecessary detail.

(31) Robotic surgical system **1000** generally includes a plurality of robot arms **1002**, **1003**; a control device **1004**; and an operating console **1005** coupled with control device **1004**. Operating console **1005** may include a display device **1006**, which may be set up in particular to display three-dimensional images; and manual input devices **1007**, **1008**, by means of which a surgeon, may be able to telemanipulate robot arms **1002**, **1003** in a first operating mode. Robotic surgical system **1000** may be configured for use on a patient **1013** lying on a patient table **1012** to be treated in a minimally invasive manner. Robotic surgical system **1000** may further include a database **1014**, in particular coupled to control device **1004**, in which are stored, for example, pre-operative data from patient **1013** and/or anatomical atlases.

(32) Each of the robot arms **1002**, **1003** may include a plurality of members, which are connected through joints, and an attaching device **1009**, **1011**, to which may be attached, for example, a surgical tool “ST” supporting an end effector **1050**, **1060**. One of the surgical tools “ST” may be ultrasonic surgical instrument **10** (FIG. **1**), wherein manual manipulation and actuation features are replaced with robotic inputs. In such configurations, robotic surgical system **1000** may include or be configured to connect to an ultrasonic generator, a power source, and cooling system. The other surgical tool “ST” may include any other suitable surgical instrument, e.g., an endoscopic camera, other surgical tool, etc. Robot arms **1002**, **1003** may be driven by electric drives, e.g., motors, that are connected to control device **1004**. Control device **1004** (e.g., a computer) may be configured to activate the motors, in particular by means of a computer program, in such a way that robot arms **1002**, **1003**, their attaching devices **1009**, **1011**, and, thus, the surgical tools “ST” execute a desired movement and/or function according to a corresponding input from manual input devices **1007**, **1008**, respectively. Control device **1004** may also be configured in such a way that it regulates the movement of robot arms **1002**, **1003** and/or of the motors.

(33) Referring to FIGS. **1**, **2**, **4**A, **4**B, waveguide **230**, as noted above, includes blade **282** disposed at a distal end thereof. Waveguide **230**, more specifically, includes a proximal body **232** (FIGS. **2** and **4**A) and a distal body **234** (FIGS. **4**A and **4**B) that includes blade **282**. In aspects, proximal body **232** and distal body **234** may be integrally formed from a single piece of material or may be separately formed and subsequently attached to one another (permanently or removably). Proximal body **232** and distal body **234** may, in aspects where proximal and distal body **232**, **234** are separately formed and subsequently attached, be attached via threaded engagement, e.g., via receipt

of a threaded plug **233** (FIG. **4**A) of proximal body **232** within a threaded bore **236** (FIGS. **4**A and **4**B) defined within a proximal end of distal body **234**. Other methods of attachment are also contemplated, such as various types of welding (e.g., inertial friction welding), brazing, soldering, diffusion bonding, etc. Proximal body **232** and/or distal body **234** may be formed from aluminum, aluminum alloy, titanium, a titanium alloy, or other suitable similar or different material(s).

(34) Proximal body **232** of waveguide **230** extends from housing **110** through at least a portion of the inner support sleeve of elongated assembly **200** to distal body **234**. Proximal body **232** is configured to operable engage ultrasonic transducer **320** such that ultrasonic motion produced by ultrasonic transducer **320** is transmitted along proximal body **232** to distal body **234** and, ultimately, blade **282** for treating tissue clamped between blade **282** and jaw **284** or positioned adjacent to blade **282**. Proximal body **232** may define a generally cylindrical-shaped configuration and may be solid, although hollow or semi-hollow configurations are also contemplated.

(35) With particular reference to FIGS. **4**A and **4**B, distal body **234** includes a base **238** and blade **282** extending distally from base **238**. Base **238** and blade **282** may be integrally formed from a single piece of material or may be separately formed and subsequently attached to one another (permanently or removably). Base **238** defines a generally cylindrical-shaped configuration, although other configurations are also contemplated. Blade **282** extends distally from base **282**. Blade **282** may define a substantially linear configuration (as shown), may define a curved configuration, or may define any other suitable configuration, e.g., straight and/or curved surfaces, portions, and/or sections; one or more convex and/or concave surfaces, portions, and/or sections; etc. With respect to curved configurations, blade **282**, more specifically, may be curved in any direction relative to jaw **284** (FIG. **1**), for example, such that the distal tip of blade **282** is curved towards jaw **284** (FIG. **1**), away from jaw **284** (FIG. **1**), or laterally (in either direction) relative to jaw **284** (FIG. **1**). Further, blade **282** may be formed to include multiple curves in similar directions, multiple curves in different directions within a single plane, and/or multiple curves in different directions in different planes. In addition, although one configuration of blade **282** is described and illustrated herein, it is contemplated that blade **282** may additionally or alternatively be formed to include any suitable features, e.g., a tapered configuration, various different cross-sectional configurations along its length, cut-outs, indents, edges, protrusions, straight surfaces, curved surfaces, angled surfaces, wide edges, narrow edges, and/or other features. In aspects, blade **282** defines a pair of relatively narrow generally convex opposed surfaces and a pair of relatively broad generally planar opposed surfaces, although other configurations are also contemplated. One of the narrow surfaces may be positioned to oppose jaw **284** (FIG. **1**), one of the broad surfaces may be positioned to oppose jaw **284** (FIG. **1**), or blade **282** and/or jaw **284** (FIG. **1**) may be relatively rotatable to position a desired surface in opposition to jaw **284** (FIG. **1**). A transition region **240** defining a gradual (or more gradual) transition between base **238** and blade **282** may also be provided.

(36) Distal body **234** may be formed from a solid material into which the various lumens detailed below are formed, although other configurations are also contemplated. Distal body **234**, more specifically, includes first and second longitudinal lumens **242**, **244** extending in substantially parallel, spaced-apart relation relative to one another through at least a portion of base **238** and blade **282**. Longitudinal lumens **242**, **244** communicate with distal apertures **243**, **245**, respectively, defined through a distal face of blade **282**. Longitudinal lumens **242**, **244** may be formed within distal body **234** during manufacture of distal body **234** (e.g., as part of an extrusion process or other suitable formation process) or may be formed subsequent to manufacture of distal body **234** (e.g., via drilling). In either configuration, longitudinal lumens **242**, **244** extend through the distal face of blade **282** proximally through blade **282** and at least a portion of base **238** to closed or open proximal ends. Alternatively, in configurations where distal body **234** is removable from proximal body **232** (FIG. **4**A), longitudinal lumens **242**, **244** may be formed within distal body **234** through a proximal end portion of base **238** (or of waveguide **230**) to or through the distal face of blade **282**.

In aspects, threaded bore **236** of base **238** communicates with open proximal ends of longitudinal lumens **242**, **244**. In such aspects, threaded engagement of threaded plug **233** of proximal body **232** (see FIG. **4**A) within threaded bore **236** of base **238** of distal body **234** to attach proximal and distal bodies **232**, **234** with one another also serves to seal the proximal ends of longitudinal lumens **242**, **244** closed (either by itself or with a seal, e.g., a grommet, o-ring, gasket, sealant, overmold, etc., disposed between proximal and distal bodies **232**, **234**, e.g., pressed distally to bottom out within threaded bore **236**). In other configurations, longitudinal lumens **242**, **244** communicate with one or more internal lumens (e.g., a single lumen or dedicated lumens corresponding to each longitudinal lumen **242**, **244**) defined within proximal body **232** and extending at least partially therethrough to closed or closable proximal ends of the internal lumen(s) of proximal body **232**. In aspects, such internal lumens of proximal body **232** extend to the proximal end of proximal body **232** and are sealed closed (themselves or with a seal) via engagement of proximal body **232** of waveguide **230** with transducer **320** (FIG. **2**), e.g., via the threaded engagement therebetween.

(37) Continuing with reference to FIGS. **4**A and **4**B, a distal transverse lumen **246** is defined at least partially through blade **282** at a distal end portion of blade **282**, e.g., within approximately 15% of a length of blade **282** from the distal end thereof, in other aspects within approximately 10%, and in still other aspects, within approximately 5%. Distal transverse lumen **246** may be formed within blade **282** via drilling, e.g., forming a distal transverse aperture **247** disposed on one side of blade **282**, or in any other suitable manner. Distal transverse lumen **246** extends transversely across one of the longitudinal lumens **242**, **244** into communication with the other longitudinal lumen **242**, **244**. In this manner, distal transverse lumen **246** establishes communication between longitudinal lumens **242**, **244** within the distal end portion of blade **282**. Distal transverse lumen **246** may not extend completely through blade **282**, e.g., such that a single distal transverse aperture **247** is formed, or may extend completely therethrough, e.g., such that a pair of opposed distal transverse apertures **247** is formed. Distal transverse lumen **246** may extend in substantially perpendicular orientation relative to longitudinal lumens **242**, **244** or at an angle relative thereto.

(38) Base **238** of distal body **234** of waveguide **230** defines a pair of proximal transverse lumens **248***a*, **248***b* therethrough, although it is also contemplated that proximal transverse lumens **248***a*, **248***b* be defined within proximal body **232** of waveguide **230** (in configuration wherein longitudinal lumens **242**, **244** communicate with corresponding lumens defined through at least a portion of proximal body **232**). Proximal transverse lumens **248***a*, **248***b* extend from transverse apertures **249***a*, **249***b* on opposing sides of base **238** into communication with longitudinal lumens **242**, **244**, respectively, or may be disposed on the same side of base **238** and extend into communication with respective longitudinal lumens **242**, **244**. Longitudinal lumens **242**, **244** may terminate at proximal transverse lumens **248***a*, **248***b* or may extend proximally therebeyond. Proximal transverse lumens **248***a*, **248***b* may be formed via drilling through base **238** or in any other suitable manner. Proximal transverse lumens **248***a*, **248***b* may extend only partially through base **238** so as to maintain isolation between longitudinal lumens **242**, **244** within base **238**, or may extend completely through base **238** whereby the undesired apertures formed thereby are sealed closed during manufacturing, e.g., according to any of the aspects detailed herein. Whether by initial formation or subsequent sealed closure, proximal transverse lumens **248***a*, **248***b* each communicate with one of the longitudinal lumens **242**, **244** but not the other longitudinal lumen **242**, **244**. Proximal transverse lumens **248***a*, **248***b* may be transversely aligned with one another or longitudinally offset relative to one another and either or both may extend in substantially perpendicular orientation relative to longitudinal lumens **242**, **244** or at angles relative thereto.

(39) Referring also to FIG. **5**, during manufacturing, in order to define a closed fluid circuit from cooling system **120**, through waveguide **230**, and back to cooling system **120** (see FIG. **2**), inflow and return conduits **128***a*, **128***b* are coupled, in fluid communication, with longitudinal lumens **242**, **244**, respectively, and the various apertures **243**, **245**, **247**, **249***a*, **249***b* defined through waveguide **230** are sealed to inhibit escape of fluid from the closed fluid circuit. Although the fluid circuit

from cooling system **120** through waveguide **230** and back to cooling system **120** is closed, cooling system **120** may itself define an open-loop configuration (e.g., wherein supply fluid and return fluid are separate and return fluid is not re-circulated), a closed-loop configuration (e.g., wherein return fluid is re-circulated as supply fluid), or a semi-closed loop configuration (e.g., wherein some return fluid is re-circulated while other return fluid is not re-circulated). Further still, in other configurations, one or more apertures **243**, **245**, **247**, **249***a*, **249***b* defined through waveguide **230** are sealed closed while one or more other apertures **243**, **245**, **247**, **249***a*, **249***b* defined through waveguide **230** remain open to enable the direction of fluid from waveguide **230** in a particular manner or manners in an open-loop system, e.g., to function as an aspirator and/or suction device.

(40) Inflow and return conduits **128***a*, **128***b* may extend to or at least partially through transverse apertures **249***a*, **249***b* to enable the delivery of inflow fluid through proximal transverse aperture **249***a*, proximal transverse lumen **248***a*, and into longitudinal lumen **242**, from longitudinal lumen **242** to longitudinal lumen **244** at the distal end portion of blade **282** via distal transverse lumen **246**, and to enable the return of return fluid from longitudinal lumen **244**, through proximal transverse lumen **248***b* and transverse aperture **249***b* to return conduit **128***b*. Inflow conduit **128***a* and/or return conduit **128***b* may, interiorly and/or exteriorly relative to waveguide **230**, be oriented substantially perpendicularly relative to longitudinal lumens **242**, **244**, respectively, or may be angled relative thereto. Additionally or alternatively, as noted above, proximal transverse lumens **248***a*, **248***b* may be angled or substantially perpendicular relative to longitudinal lumens **242**, **244**. Angling inflow conduit **128***a* and/or return conduit **128***b* and/or angling either or both proximal transverse lumens **248***a*, e.g., in a proximally-angled manner, may facilitate inflow and outflow of fluid to and from longitudinal lumens **242**, **244**, respectively, although other configurations are also contemplated. In configurations where inflow and return conduits **128***a*, **128***b* extend at least partially into proximal transverse lumens **248***a*, **248***b*, respectively, the ends of inflow and return conduits **128***a*, **128***b* may be cut at an angled, chamfered, or otherwise asymmetrically configured to facilitate fluid flow in a desired manner, e.g., distally from inflow conduit **128***a* through longitudinal lumen **242** and/or proximally through longitudinal lumen **244** into return conduit **128***b*.

(41) Referring still to FIGS. **4**A-**5**, apertures **243**, **245**, **247**, **249***a*, **249***b* may be closed via corresponding plugs **260**. Plugs **260** may be similar or different from one another and may sealingly close apertures **243**, **245**, **247**, **249***a*, **249***b* in similar or different manners. One or more of plugs **260** may be configured, for example, as a rod, e.g., a titanium rod or other rod of similar of different material from distal body **234** of waveguide **230**, that is inserted into the corresponding aperture **243**, **245**, **247**, **249***a*, **249***b* and welded therein to sealingly close the aperture **243**, **245**, **247**, **249***a*, **249***b*. In such aspects, lost wax, depth gauge, or other suitable fixturing may be utilized to retain the rod in pace during welding. In aspects, one or more of plugs **260** may be formed from, for example, a high temperature polymer, rubber, or metal and, in these or other aspects, may be welded, thermally interference fit, soldered (wherein the plugs are formed wholly from the solder material or wherein a plug of different material is soldered), brazed, or otherwise secured in position within the corresponding aperture **243**, **245**, **247**, **249***a*, **249***b* to sealingly close the corresponding aperture **243**, **245**, **247**, **249***a*, **249***b*.

(42) The plug **260** configured to sealingly close apertures **243**, **245**, in aspects, may be joined via a connector such that plugs **260** are together inserted into corresponding apertures **243**, **245** and secured therein (in any manner detailed herein or any other suitable manner, and with plugs **260** formed from any suitable material such as any of those detailed herein). Such a plug **260** may define a substantially U-shaped configuration and be inserted from the exterior of waveguide **230**. Alternatively, such a plug **260** may include an O-ring (itself or together with plug portions extending therefrom) or other seal connector, e.g., a gasket, that is inserted distally through lumen **242**, **244** to seal apertures **243**, **245** (and, in some aspects, also aperture **247**) closed.

(43) One or more of plugs **260** may be overmolded from the interior or exterior of the corresponding aperture **243**, **245**, **247**, **249***a*, **249***b* to sealingly close the corresponding aperture

243, **245**, **247**, **249***a*, **249***b*. One or more of plugs **260** may be separately formed and then inserted (from the interior or exterior) and sealed within the corresponding aperture **243**, **245**, **247**, **249***a*, **249***b*, or may be formed within the corresponding aperture **243**, **245**, **247**, **249***a*, **249***b*, e.g., via filling the corresponding aperture with material via overmolding, casting, brazing, soldering, etc. from the interior or exterior thereof. One or more of plugs **260** may be configured as a set screw or other threaded element and/or one or more of apertures **243**, **245**, **247**, **249***a*, **249***b* may include threading to enable sealed engagement via threading (itself (e.g., via pipe thread sealing) or with an additional seal or seal material) of the one or more plugs **260** within the corresponding aperture(s) **243**, **245**, **247**, **249***a*, **249***b*, from the interior or exterior thereof.

(44) In configurations wherein longitudinal lumens **242**, **244** define open proximal ends, the open proximal ends may be collectively or individually plugged according to any of the aspects detailed above or in any other suitable manner. Further, with respect to the plugs **260** for apertures **249***a*, **249***b*, inflow and outflow conduits **128***a*, **128***b*, respectively, may extend through the plugs **260**, be defined within the plugs **260**, form the plugs **260**, be surrounded by the plugs **260**, or be otherwise configured relative to the plugs **260** such that the plugs **260** seal apertures **249***a*, **249***b* and inflow and outflow conduits **128***a*, **128***b*, respectively, to inhibit escape of fluid from the system while allowing fluid flow between waveguide **230** and inflow and outflow conduits **128***a*, **128***b*. Combinations of any two or more of the above-noted plug configurations are also contemplated as are other configurations to facilitate sealed closure of some or all apertures **243**, **245**, **247**, **249***a*, **249***b*.

(45) With general reference to FIGS. **1-5**, in use, upon activation of cooling, cooling system **120** is configured to pump fluid, using the one or more fluid pumps **122**, from fluid reservoir **124**, through inflow conduit **128***a*, proximal transverse lumen **248***a*, distally through longitudinal lumen **242**, from longitudinal lumen **242** to longitudinal lumen **244** at the distal end portion of blade **282** via distal transverse lumen **246**, proximally through longitudinal lumen **244**, through transverse lumen **248***b*, through outflow conduit **128***b*, and back to fluid reservoir **124**. In this manner, cooling fluid is circulated substantially entirely along the length of blade **282** to facilitate cooling blade **282**. The cooling fluid may be saline, water, or other suitable fluid.

(46) Turning to FIGS. **6**A-**6**B, a distal body **1234** including a blade **1282** of another waveguide **1230** provided in accordance with aspects of the present disclosure is shown. Except as explicitly contradicted below, waveguide **1230** may be configured similarly to and include any of the features of waveguide **230** (FIGS. **1**, **2**, **4**A, and **4**B). Thus, the following description will focus on differences between waveguide **1230** and waveguide **230** (FIGS. **1**, **2**, **4**A, and **4**B) while similarities are only summarily described or omitted entirely.

(47) Distal body **1234** of waveguide **1230** includes a base **1238** and blade **1282** extending distally from base **1238**. Distal body **1234** further includes first and second longitudinal lumens **1242**, **1244** extending in substantially parallel, spaced-apart relation relative to one another through at least a portion of base **1238** and blade **1282** to open distal ends. A cap **1250** defining a connector lumen **1252** is welded, soldered, or otherwise secured to the distal end of blade **1282** to thereby define a distal tip of blade **1282**. Cap **1250** may be formed from the same material as blade **1282** or a different material. Upon securing cap **1250** to the distal end of blade **1282**, connector lumen **1252** establishes communication between the open distal ends of longitudinal lumens **1242**, **1244** and otherwise seals off the fluid flow path, inhibiting the escape of fluid from within distal body **1234**. Thus, cooling fluid can be pumped distally through distal body **1234** via one of longitudinal lumens **1242**, **1244** and can return proximally through distal body **1234** via connector lumen **1252** and the other longitudinal lumen **1242**, **1244** to thereby cool distal body **1234** of waveguide **1230**. With respect to distal body **1234**, a transverse lumen need not be provided towards distal ends of longitudinal lumens **1242**, **1244**.

(48) With reference to FIG. **7**, a distal body **2234** including a blade **2282** of another waveguide **2230** provided in accordance with aspects of the present disclosure is shown. Distal body **2234** is

similar to and may include any of the features of distal body **1234** (FIGS. **6**A and **6**B) except as explicitly contradicted below.
(49) Distal body **2234** of waveguide **2230** includes a base **2238** and blade **2282** extending distally from base **2238**. Distal body **2234** further includes first and second longitudinal lumens **2242**, **2244** extending in substantially parallel, spaced-apart relation relative to one another through at least a portion of base **2238** and blade **2282**. Longitudinal lumens **2242**, **2244** terminate at and communicate with a transverse lumen **2246** disposed at the distal end of blade **2282** and in communication with an open distal end thereof. Thus, lumens **2242**, **2244**, **2246** all communicate with the open distal end of blade **2282**.
(50) A cap **2250** including a base **2252** and a head **2254** is welded, soldered, or otherwise secured to the distal end of blade **2282** to thereby define a distal tip of blade **2282** with base **2252** extending at least partially into the open distal end of blade **2282** and head **2254** abutting the distal face(s) of blade **2282**. Upon securing cap **2250** to the distal end of blade **2282** in this manner, the open distal end of blade **2282** is sealed closed while still maintaining a fluid flow path between longitudinal lumens **2242**, **2244** via transverse lumen **2246**. Thus, cooling fluid can be pumped distally through distal body **2234** via one of longitudinal lumens **2242**, **2244** and can return proximally through distal body **2234** via transverse lumen **2246** and the other longitudinal lumen **2242**, **2244** to thereby cool distal body **2234** of waveguide **2230**. Base **2252** of cap **2250** facilitates proper alignment and orientation of cap **2250** relative to blade **2282** upon welding or otherwise securing cap **2250** to blade **2282**.
(51) Referring to FIGS. **8**A and **8**B, a distal body **3234** including a blade **3282** of another waveguide **3230** provided in accordance with aspects of the present disclosure is shown. Distal body **3234** is similar to and may include any of the features of distal body **1234** (FIGS. **6**A and **6**B) except as explicitly contradicted below.
(52) Rather than providing a cap **1250** as detailed above with respect to distal body **1234** (see FIGS. **6**A and **6**B), distal body **3234** includes a bent pipe **3250** (curved, angled, and/or otherwise bent) defining a lumen **3252** therethrough secured, e.g., welded, soldered, or otherwise secured, to the distal end of blade **3282** such that lumen **3252** establishes communication between the open distal ends of longitudinal lumens **3242**, **3244** and otherwise seals off the fluid flow path, inhibiting the escape of fluid from distal body **3234**. Thus, cooling fluid can be pumped distally through distal body **3234** via one of longitudinal lumens **3242**, **3244** and can return proximally through distal body **3234** via lumen **3252** and the other longitudinal lumen **3242**, **3244** to thereby cool distal body **3234** of waveguide **3230**.
(53) With general reference to FIGS. **6**A-**8**B, in aspects, cap **1250**, cap **2250**, and/or pipe **3250** may define features to facilitate tissue treatment. For example, cap **1250**, cap **2250**, and/or pipe **3250** may define angled, rounded, curved, and/or other suitable configurations to facilitate, for example, performing otomies, blunt dissection, and/or other surgical tasks with or without ultrasonic energy.
(54) Turning to FIGS. **9** and **10**, proximal end portions of the distal bodies **4234**, **5234** of still other waveguides **4230**, **5230** provided in accordance with aspects of the present disclosure are shown. Except as explicitly contradicted below, waveguides **4230**, **5230** may be configured similarly to and include any of the features of waveguide **230** (FIGS. **1**, **2**, **4**A, and **4**B), any of the other waveguides detailed herein, and/or any other suitable waveguide. Thus, the following description will focus on differences between waveguides **4230**, **5230** and the above-detailed waveguides, e.g., waveguide **230** (FIGS. **1**, **2**, **4**A, and **4**B), while similarities are only summarily described or omitted entirely.
(55) Referring to FIG. **9**, distal body **4234** of waveguide **4230** defines longitudinal lumens **4242**, **4244** extending at least partially therethrough. Longitudinal lumens **4242**, **4244** define open proximal ends open to the proximal end of distal body **4234**. Distal body **4234** of waveguide **4230** further includes a connector body **4250** secured, e.g., welded, soldered, or otherwise secured, to the proximal end of distal body **4234**. Connector body **4250** may be formed from the same or a

different material as distal body **4234** and, upon securing connector body **4250** to the proximal end of distal body **4234**, seals closed the proximal ends of longitudinal lumens **4242**, **4244**. Connector body **4250** may further include a proximal bore **4236** to enable threaded or other suitable engagement of the proximal body (not shown, see proximal body **232** (FIG. **4**A)) of waveguide **4230** within proximal bore **4236** to thereby secure the proximal body and distal body **4234** with one another (via connector body **4250** therebetween) to enable ultrasonic energy transmission therealong.

(56) With reference to FIG. **10**, distal body **5234** of waveguide **5230** defines longitudinal lumens **5242**, **5244** extending at least partially therethrough and a proximal bore **5236** defined therein at the proximal end thereof. The proximal ends of longitudinal lumens **5242**, **5244** communicate with proximal bore **5236**. A plug **5250** is secured within proximal bore **5236** at the open proximal ends of longitudinal lumens **5242**, **5244** to seal off the proximal ends of longitudinal lumens **5242**, **5244**. Plug **5250** may be formed from any suitable material similar or different from that of waveguide **5230** and may be welded, soldered, adhered, molded, compression-fit (such as where plug **5250** is formed from a resilient material), threaded, or otherwise secured within proximal bore **5236**. Plug **5250** occupies only a portion of proximal bore **5236** to enable threaded or other suitable engagement of the proximal body (not shown, see proximal body **232** (FIG. **4**A)) of waveguide **5230** within proximal bore **5236** to thereby secure the proximal body and distal body **5234** with one another to enable ultrasonic energy transmission therealong.

(57) While several aspects and features of the disclosure are detailed above and shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Therefore, the above description and accompanying drawings should not be construed as limiting, but merely as exemplifications of particular configurations. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

Claims

1. An ultrasonic surgical system, comprising: an ultrasonic waveguide body comprising: a proximal waveguide body adapted to connect to an ultrasonic transducer, and a distal waveguide body including a blade and defining first and second longitudinal lumens extending through at least a portion of a length of the distal waveguide body, the distal waveguide body extending distally from the proximal waveguide body; a distal transverse lumen that intersects and interconnects the first and second longitudinal lumens; a first proximal transverse lumen extending from a first proximal transverse aperture within the ultrasonic waveguide body transversely through a portion of the ultrasonic waveguide body to intersect the first longitudinal lumen, wherein the first proximal transverse aperture is in a first side of the ultrasonic waveguide body; a second proximal transverse lumen extending from a second proximal transverse aperture within the ultrasonic waveguide body transversely through a portion of the ultrasonic waveguide body to intersect the second longitudinal lumen, wherein the second proximal transverse aperture is in a second side of the ultrasonic waveguide body, and the second proximal transverse aperture is transversely aligned with the first proximal transverse aperture; and inflow and outflow conduits fluidly coupled with the first and second proximal transverse apertures, respectively, to enable inflow of fluid into the first longitudinal lumen and outflow of fluid from the second longitudinal lumen, respectively, wherein the distal waveguide body is releasably engageable with the proximal waveguide body.
2. The ultrasonic surgical system according to claim 1, wherein first and second proximal transverse aperture plugs form a seal between the inflow and outflow conduits and the first and second proximal transverse apertures, respectively.
3. The ultrasonic surgical system according to claim 1, wherein the distal waveguide body includes a base and a blade extending distally from the base, the blade defining opposed narrow surfaces

and opposed broad surfaces.

4. The ultrasonic surgical system according to claim 3, wherein at least a portion of the distal transverse lumen is disposed within 10% of a length of the blade from a distal end of the blade.

5. The ultrasonic surgical system according to claim 1, further comprising: a cooling system configured to pump cooling fluid through the inflow conduit into the first longitudinal lumen and/or pump cooling fluid through the second longitudinal lumen into the outflow conduit; a housing at least partially supporting the cooling system; and an elongated assembly extending distally from the housing, the elongated assembly including the ultrasonic waveguide body.

6. The ultrasonic surgical system according to claim 5, wherein the housing further supports an ultrasonic transducer configured to produce ultrasonic energy for transmission along the ultrasonic waveguide body.

7. The ultrasonic surgical system according to claim 6, wherein the housing further supports an ultrasonic generator configured to produce an ultrasonic drive signal for driving the ultrasonic transducer, and a battery configured to power the ultrasonic generator.

8. An ultrasonic surgical system, comprising: an ultrasonic waveguide body defining first and second longitudinal lumens extending through at least a portion of a length of the ultrasonic waveguide body; a distal transverse lumen defined within the ultrasonic waveguide body transversely through at least a portion of the ultrasonic waveguide body to intersect and interconnect the first and second longitudinal lumens, wherein the distal transverse lumen extends from a distal transverse aperture and wherein a distal transverse aperture plug closes the distal transverse aperture to inhibit escape of fluid therethrough; a first proximal transverse lumen extending from a first proximal transverse aperture within the ultrasonic waveguide body transversely through a portion of the ultrasonic waveguide body to intersect the first longitudinal lumen; a second proximal transverse lumen extending from a second proximal transverse aperture within the ultrasonic waveguide body transversely through a portion of the ultrasonic waveguide body to intersect the second longitudinal lumen; and inflow and outflow conduits fluidly coupled with the first and second proximal transverse apertures, respectively, to enable inflow of fluid into the first longitudinal lumen and outflow of fluid from the second longitudinal lumen, respectively.

9. The ultrasonic surgical system according to claim 8, wherein the first and second longitudinal lumens extend proximally from first and second distal face apertures defined within a distal face of the ultrasonic waveguide body, and wherein at least one distal face aperture plug closes the first and second distal face apertures to inhibit escape of fluid therethrough.

10. An ultrasonic surgical system, comprising: a proximal waveguide body adapted to connect to an ultrasonic transducer; a distal waveguide body including a blade and defining first and second longitudinal lumens extending through at least a portion of a length of the distal waveguide body, the distal waveguide body extending distally from the proximal waveguide body; a distal transverse lumen defined within the distal waveguide body transversely through at least a portion of the distal waveguide body to intersect and interconnect the first and second longitudinal lumens; a first proximal transverse lumen extending from a first proximal transverse aperture within the distal waveguide body transversely through a portion of the distal waveguide body to intersect the first longitudinal lumen; a second proximal transverse lumen extending from a second proximal transverse aperture within the distal waveguide body transversely through a portion of the distal waveguide body to intersect the second longitudinal lumen; and inflow and outflow conduits fluidly coupled with the first and second proximal transverse apertures, respectively, to enable inflow of fluid into the first longitudinal lumen and outflow of fluid from the second longitudinal lumen, respectively, wherein the distal waveguide body is releasably engageable with the proximal waveguide body.

11. The ultrasonic surgical system according to claim 10, wherein engagement of the proximal and distal waveguide bodies closes proximal ends of the first and second longitudinal lumens.

12. An ultrasonic surgical system, comprising: an ultrasonic waveguide body defining first and

second longitudinal lumens extending through at least a portion of a length of the ultrasonic waveguide body; a distal transverse lumen defined within the ultrasonic waveguide body transversely through at least a portion of the ultrasonic waveguide body to intersect and interconnect the first and second longitudinal lumens, wherein the distal transverse lumen is at least partially formed via a distal cap defining a distal tip of the ultrasonic waveguide body; a first proximal transverse lumen extending from a first proximal transverse aperture within the ultrasonic waveguide body transversely through a portion of the ultrasonic waveguide body to intersect the first longitudinal lumen; a second proximal transverse lumen extending from a second proximal transverse aperture within the ultrasonic waveguide body transversely through a portion of the ultrasonic waveguide body to intersect the second longitudinal lumen; and inflow and outflow conduits fluidly coupled with the first and second proximal transverse apertures, respectively, to enable the inflow of fluid into the first longitudinal lumen and the outflow of fluid from the second longitudinal lumen, respectively.

13. A method of manufacturing an ultrasonic surgical system, comprising: forming first and second longitudinal lumens through at least a portion of a length of an ultrasonic waveguide body; forming a distal transverse lumen transversely through at least a portion of the ultrasonic waveguide body to intersect and interconnect the first and second longitudinal lumens; forming first and second proximal transverse lumens transversely through a portion of the ultrasonic waveguide body to intersect the first and second longitudinal lumens, respectively; plugging apertures formed from the formation of the first and second longitudinal lumens; and plugging an aperture formed from the formation of the distal transverse lumen.

14. The method according to claim 13, further comprising fluidly connecting an inflow conduit with the first proximal transverse aperture.

15. The method according to claim 13, further comprising fluidly connecting an outflow conduit with the second proximal transverse aperture.

16. The method according to claim 13, further comprising attaching the ultrasonic waveguide body, as a distal waveguide body, to a proximal waveguide body.

17. The method according to claim 16, wherein the attaching closes proximal ends of the first and second longitudinal lumens.

18. The method according to claim 13, further comprising: coupling the first and second proximal transverse lumens with inflow and outflow conduits, respectively, associated with a cooling system including at least one pump and a fluid reservoir.

19. The method according to claim 13, wherein forming the distal transverse lumen includes securing a cap to a distal end of the ultrasonic waveguide body to define a distal tip thereof.